

The Mathematical Blind Spot in Responsive Images

Architecturally, why `srcset` using `w` descriptors is fundamentally crippled without the `sizes` attribute during the browser's preload scanning phase.

🔍 Browser Preload Scanner (Early Parse Phase)

🖋️ CSSOM / Layout

```
<!-- HTML arriving over network stream -->

<!-- FATAL FLAW: No 'sizes' defined! -->
```

```
img {
  width: 33vw;
}
```

🚫 BLOCKED

📦 Known Variables

📏 Intrinsic Widths (W_{img})
[400, 800, 1200]

400w 800w 1200w

🖥️ Viewport (V_w)
1000px

🔍 Device Pixel Ratio (DPR)
2.0x (High Density)

📐 The Selection Equation

To calculate required intrinsic resolution (W_{req}):

$$W_{req} = W_{layout} \times DPR$$

$$W_{req} = ??? \times 2.0$$

❓ Unknown: W_{layout}

Cannot Access! 🚫

The Preload Scanner is decoupled from CSS layout. It cannot wait for CSS to download and parse.

⚠️ Layout width is trapped!

🧠 Architectural Consequence: The Default Guess

Because the browser has been blinded to the actual layout size (W_{layout}), it mathematically must assume the worst-case scenario to avoid serving a blurry image. It assumes $W_{layout} = 100vw$ (the entire viewport).

$$\text{Calculation: } W_{req} = 1000px \times 2.0 = 2000px.$$

The browser downloads `large.jpg` (1200w). However, the CSS eventually renders the image at **33vw (330px)**.

❗ Result: Wasted bandwidth, memory overhead, and severe performance degradation!